

Surgery for bone cancer

Surgery is one of the main treatments for primary bone cancer. Surgery involves removing all or part of the cancer with an operation.

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About bone cancer surgery

Surgery is usually the main treatment for most types of [primary bone cancer](#). When surgery is not possible, your cancer doctor will talk to you about other treatments.

Primary bone cancers are rare, so surgery is done by a specialist surgeon as part of a bone sarcoma [multidisciplinary team \(MDT\)](#). They are based in a specialist sarcoma unit.

Types of surgery for bone cancer

The type of operation depends on where the cancer is in the body.

Some operations for primary bone cancer are minor. Others may involve major surgery and could affect how that area of the body works or looks.

Sometimes reconstructive surgery is also needed so the area affected can work or move as well as possible. In some cases, the surgeon can remove the affected area of bone without replacing it.

Wide local excision

The aim of most sarcoma surgery is to remove (excise) all of the cancer from the bone. Surgery may affect the way the body part moves or works, but surgeons will try to keep as much movement as possible. Usually, they do an operation called a wide local excision. This means the surgeon removes the cancer, along with an area (margin) of healthy, cancer-free tissue around it.

After the operation, the pathologist looks at the tissue from the margin under a microscope. If there are cancer cells in the margin, you may need another operation to remove more tissue. Making sure the margins are clear reduces the risk of the cancer coming back.

Most bone cancers affect the [arms and legs \(limbs\) or pelvis](#). Surgery to these areas is done by a bone surgeon called an orthopaedic surgeon. If the cancer is in the bones of the face, skull or chest, a surgeon who specialises in those areas will do the surgery.

Your surgeon will talk with you about which operation they think is best for you.

Surgery to the arms and legs (limbs)

It is usually possible to remove a primary bone cancer in an arm or leg without having to remove the whole limb. This is called [limb-sparing or limb-salvage surgery](#). You will usually have [chemotherapy](#) before and after the operation. Sometimes [radiotherapy](#) may also be used.

Rarely, the most effective treatment option is to remove the affected limb. This is called an [amputation](#). This is to make sure all the cancer has been removed. It may also be done if the surgeon thinks that limb sparing surgery may cause problems with movement of the limb.

Surgery to other areas of the body

Bone cancer in the pelvis

When possible, a wide local excision is used to remove primary bone cancer in the pelvis. Depending on the type of bone cancer, [chemotherapy](#) may be used before the operation. This is to help shrink the cancer before having the surgery. For some types of bone cancer, chemotherapy is sometimes used after surgery.

Radiotherapy may also be given before or after surgery.

Sometimes, the pelvis is reconstructed using bone taken from another area of the body. This is called a bone graft.

If lymph nodes in the pelvis are removed, this can increase the risk of [lymphoedema](#) in your legs. Lymphoedema is swelling near to where the lymph nodes were removed. Your doctor and nurse will explain more about this.

Bone cancer in the jawbone

The whole bone is usually removed and then reconstructed using bone taken from another part of the body. If it is not possible to remove the whole bone, [radiotherapy](#) may also be used.

Bone cancer in the spine or skull

If it is possible to remove all the cancer safely, a combination of treatments might be used. These include surgery, [chemotherapy](#) and [radiotherapy](#).

Reconstructive surgery

Depending on the size and position of the bone cancer, the surgeon may need to remove a large area of bone.

This means you may have surgery to reconstruct the area at the same time as your operation. It may be possible to replace the bone that is removed with a metal implant. This is called an endoprosthetic replacement (EPR).

If a metal implant or a bone graft is needed, surgery is done by a [plastic surgeon and an orthopaedic \(bone\) surgeon](#).

The aim of reconstructive surgery is to make the area look and work as naturally as possible. The surgery you have depends on:

- where in your body the cancer is
- how much bone and tissue the surgeon needs to remove.

Rehabilitation can take time. You will have support with your recovery and go to the hospital gym for physiotherapy. This will help you recover and reduce the risk of problems after surgery.

Your surgeon and specialist nurse will talk to you about the operation. They can answer any questions you have and give you an idea of how the area will look after surgery. They may be able to show you photographs of others who have had the same surgery.

Body image after surgery to the bone

Having surgery to the bone can affect how you think and feel about [your body \(body image\)](#). Your doctor or nurse can arrange for you to see a [psychologist or counsellor](#) if you feel you need more support with this.

Surgery to remove secondary cancer

Surgery is sometimes used to remove a primary bone cancer that has:

- spread to other parts of the body (secondary cancer)
- come back after treatment (recurrence).

It is mainly done when the cancer has spread to the lungs. An operation may only be possible if the secondary cancers are small and there are only a few of them.

Sometimes surgery is used for cancer in other parts of the body. If you need to have this operation, your cancer doctor will talk with you about it.

[Chemotherapy](#) may be given first. This can help shrink the secondary cancers and help make an operation possible.

Before and after surgery for bone cancer

Before surgery your surgeon and specialist nurse will tell you about the side effects and possible risks. It is important you understand what your operation may involve.

Before surgery, you may be given advice on:

- stopping smoking to help you recover and reduce your risk of problems after surgery.
- exercises to help you get back to your usual activities after surgery, or to get fitter - for example, if you will use crutches or a walking frame after surgery, they might suggest upper body strength exercises.
- eating healthily or taking supplement drinks.

Depending on the type of surgery you have, may be referred to a rehabilitation team. They will help you recover from the operation, and make sure the area being operated on moves and works as well as possible.

About our information

References



Below is a sample of the sources used in our information about primary bone cancer. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

European Society for Medical Oncology, ESMO. 2021. Bone sarcomas: ESMOeEURACANeGENTURISeERN PaedCan Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology. S. J. Strauss¹ et al. Available at: <https://www.esmo.org/guidelines/guidelines-by-topic/sarcoma-and-gist/bone-sarcomas> (accessed July 2023)

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Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by senior medical editor [Fiona Cowie](#), Consultant Clinical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 August 2022 | Next review: 01 August 2025

⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help